

Heart Disease Analysis — Comprehensive Project Report

Group 8

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Summary/Abstract

This project analyzes the Behavioral Risk Factor Surveillance System (BRFSS) dataset to understand the prevalence and predictors of heart disease among over 319,000 respondents. The analysis explores demographic, lifestyle, dietary, and clinical risk factors while developing dashboards and visual interpretations to identify actionable insights. The project aims to understand how variables such as age, sleep, mental health, physical activity, and stroke history correlate with heart disease. Findings indicate that age, stroke, general health, and physical inactivity are strong predictors, while dietary habits alone do not significantly reduce risk. The study provides a foundation for predictive modeling and public health recommendations.

Introduction

Heart disease continues to be one of the leading global causes of mortality. The increasing availability of large-scale health data allows researchers to study contributing factors and identify high-risk population groups. The BRFSS dataset provides detailed information about lifestyle, mental and physical health, and demographic characteristics, enabling comprehensive exploration of heart disease patterns.

Motivation:

- Rising global burden of cardiovascular diseases
- Availability of large, high-quality public health datasets
- Need for evidence-based public health interventions

Objectives:

- Identify key demographic, lifestyle, and clinical variables associated with heart disease
- Develop dashboards to visualize risk patterns
- Conduct exploratory data analysis (EDA) to understand relationships
- Provide insights that can support health awareness and policy decisions

Methodology

Data Source:

Source: CDC Behavioral Risk Factor Surveillance System (BRFSS)

Total Records: 319,795 respondents

Target Variable: HeartDisease (Prevalence: 8.56%)

Dataset Dimensions:

- Demographics: Age Category, Sex, Race, BMI
- Lifestyle: Smoking, Physical Activity, Sleep Time
- Clinical History: Stroke, Diabetes
- Well-being: General Health, Poor Physical Health Days, Poor Mental Health Days
- Dietary Patterns: Fruit Consumption, Fried Potato Consumption

The dataset offers valuable insights into population health, risk factors, lifestyle behaviors, and potential predictors of heart disease.

1. Data Understanding:

- Reviewing the dataset structure and variable definitions
- Identifying missing values and inconsistencies
- Understanding categorical and numerical variable distributions

2. Data Cleaning:

- Removing duplicates
- Handling missing data
- Converting categorical variables

- Standardizing variable formats
- Outlier checks

3. Exploratory Data Analysis (EDA):

- Univariate, bivariate, and multivariate exploration
- Visual identification of patterns and correlations

4. Dashboard Planning:

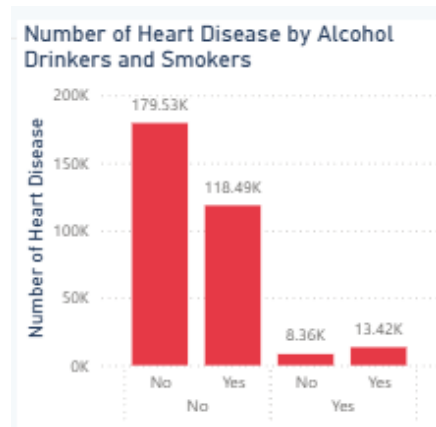
- Defining visual themes
- Linking research questions to meaningful visuals

5. Final Report & Presentation:

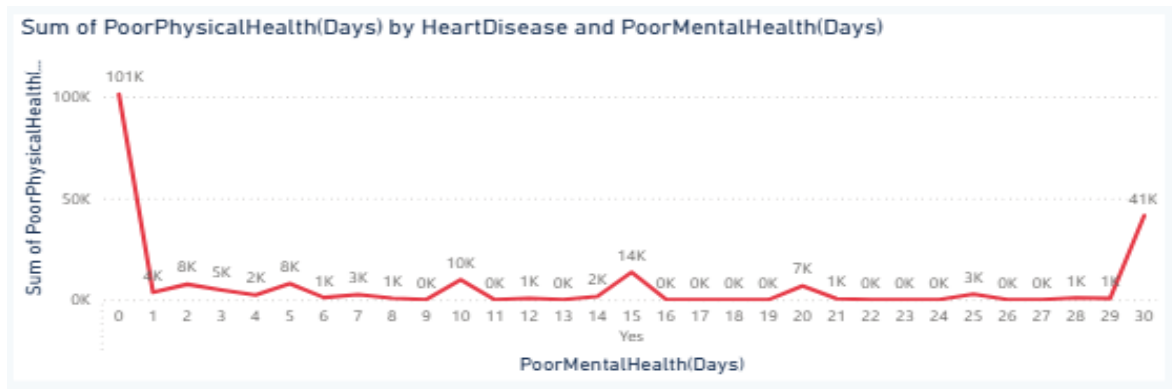
- Structuring findings
- Presenting insights with clarity

Research Questions

1. Which lifestyle factors are most strongly associated with heart disease?

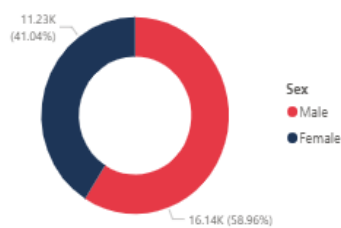


2. Do mental and physical health (poor health days) predict heart disease likelihood?

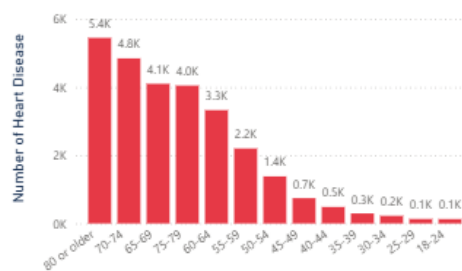


3. How do demographic factors (Age, Sex, Race) influence heart disease prevalence?

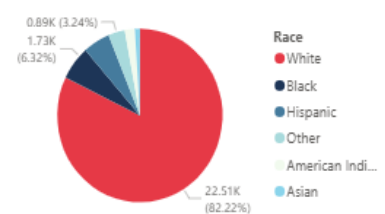
Number of Heart Disease by Sex



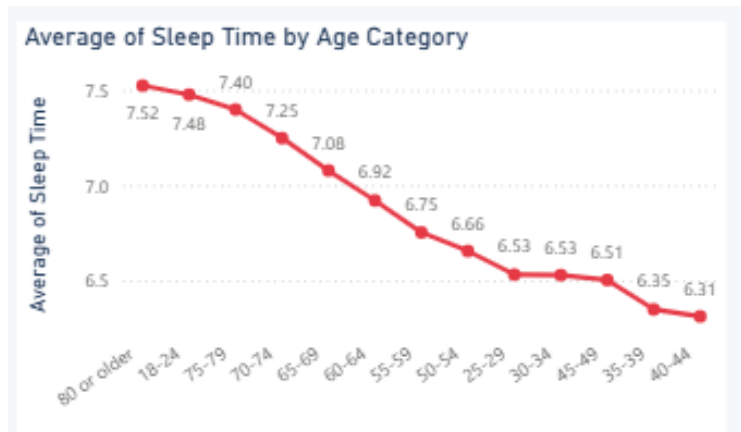
Number of Heart Disease by Age Category



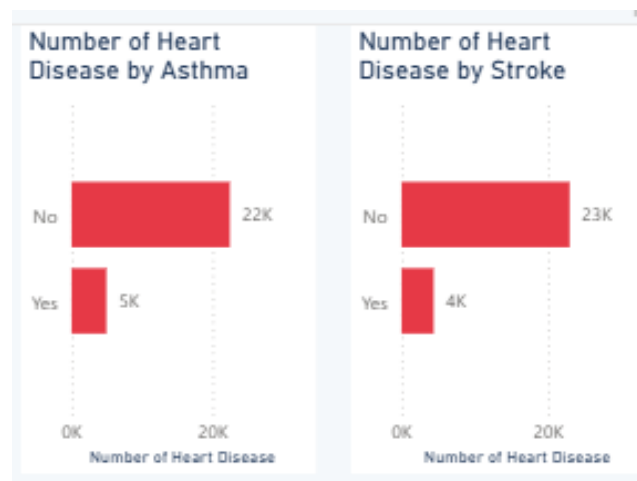
Percentage of Heart disease by Race



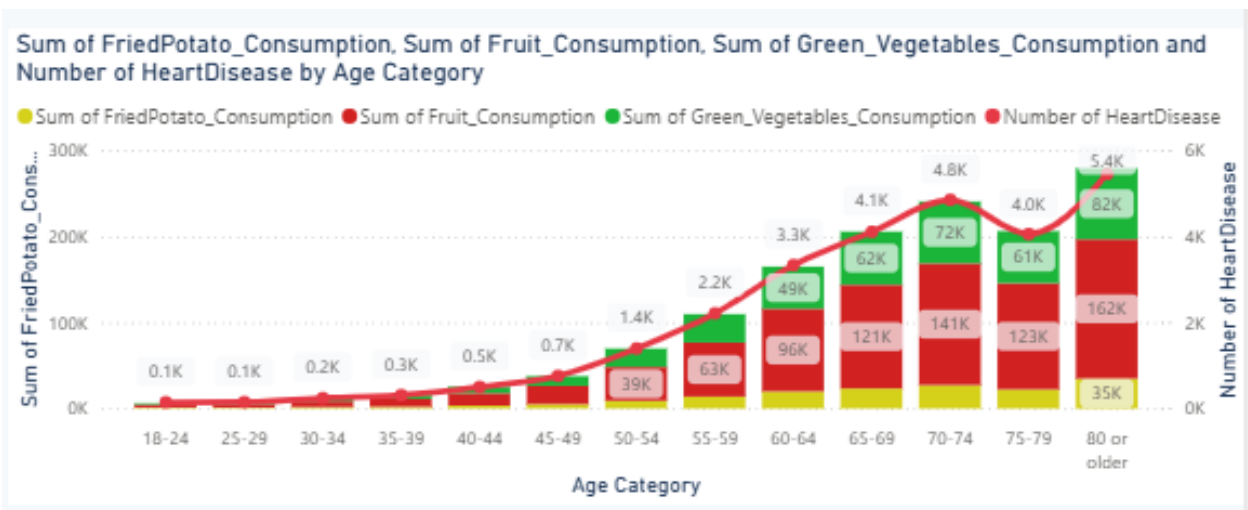
4. Does sleep duration impact chances of heart disease?



5. Does prior stroke predict heart disease risk?



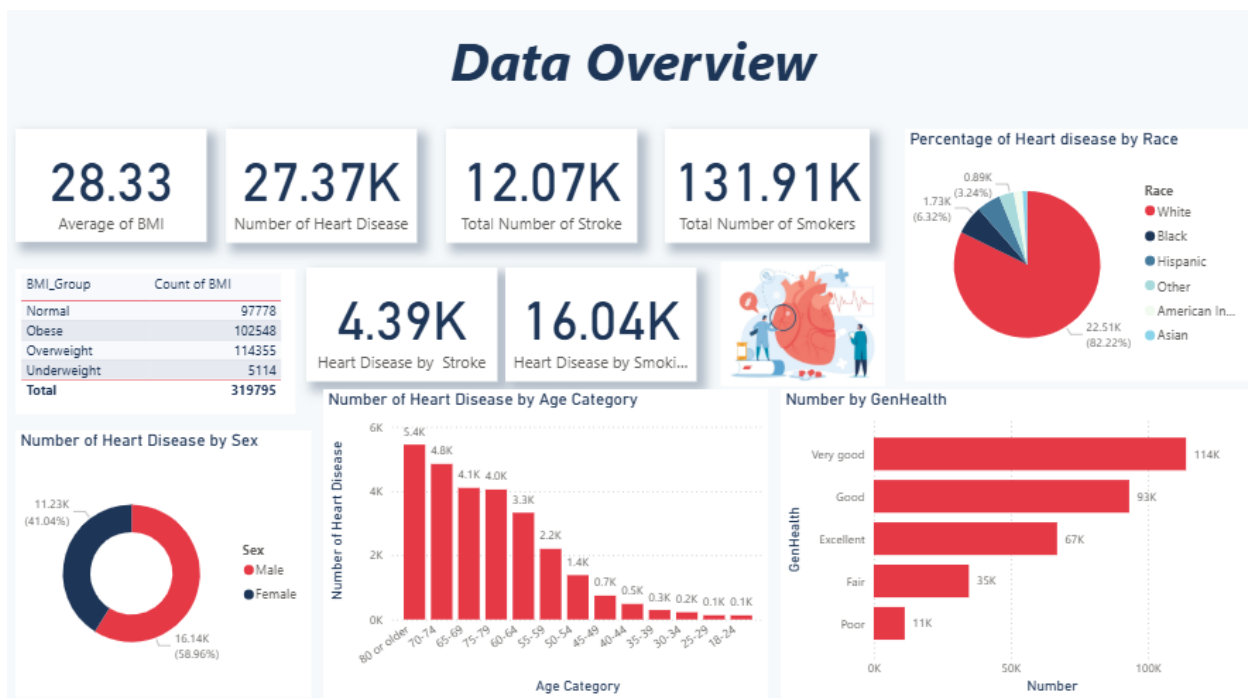
6. Is there a connection between dietary habits and heart disease?



Dashboard and Visual Analysis

Dashboard 1: Lifestyle Factors

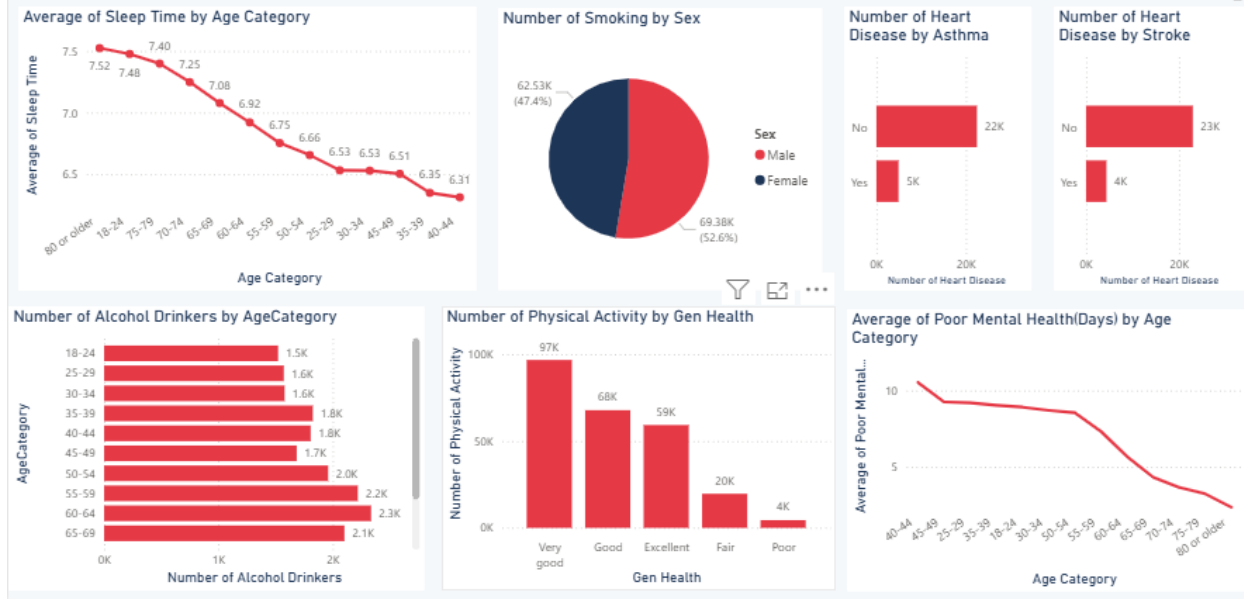
- Examines smoking, physical activity, and sleep time.
- Smokers show higher heart disease prevalence.
- Physically inactive individuals report significantly poorer general health.



Dashboard 2: Demographic Overview

- Age is one of the strongest demographic indicators.
- Heart disease prevalence rises significantly after age 55.
- Males report higher prevalence than females.
- White individuals show higher rates compared to other races.

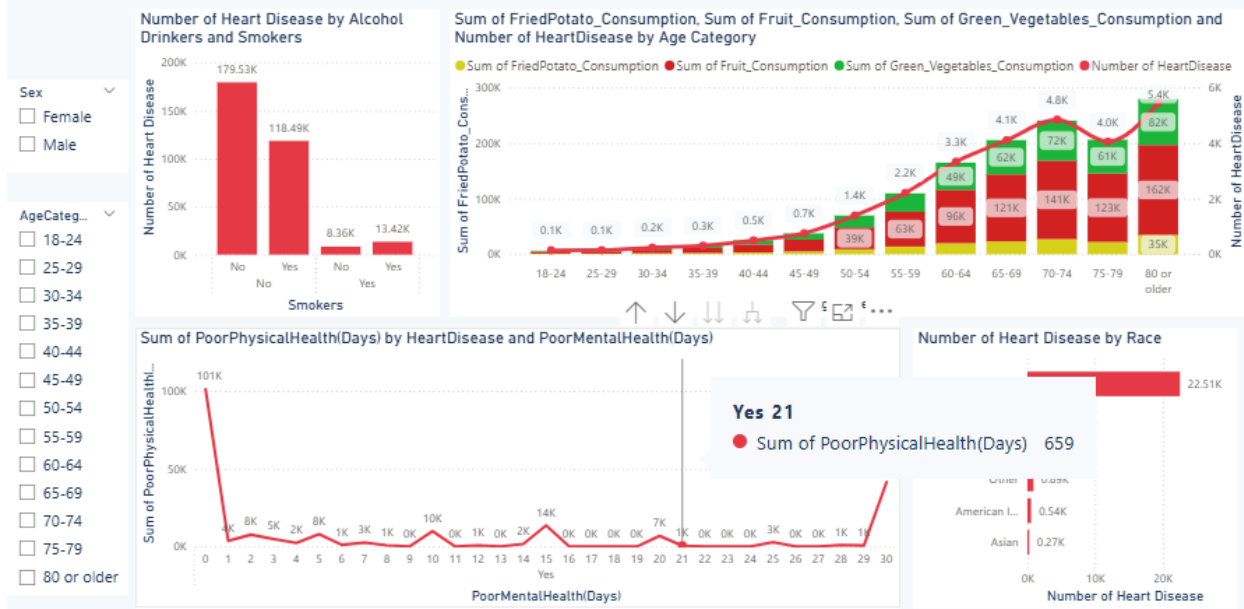
Health & Lifestyle Impact



Dashboard 3: Diet and Health Patterns

- Older adults consume more fruits/vegetables, yet still show higher heart disease prevalence.
- This suggests diet alone cannot offset age-related physiological risks.

Heart Disease Analysis



Recommendations and Future work

Recommendations:

- Promote targeted health awareness for high-risk groups (age 55+, smokers, physically inactive).
- Encourage regular physical activity programs.
- Improve public health messaging around sleep quality and duration.
- Expand mental health support, given its indirect impact on heart disease.
- Implement screening programs for individuals with prior stroke or poor general health.

Future Work:

- Apply machine learning models (Random Forest, Logistic Regression, XGBoost) to identify the strongest predictors.
- Build a prediction dashboard integrating risk scoring.
- Incorporate additional datasets to improve accuracy.
- Analyze deeper dietary patterns using more granular data.

Challenges Faced

Two primary challenges shaped this project:

1. Dataset Selection:

- Many public health datasets lacked the required variables.
- Several were behind paywalls or lacked documentation.
- BRFSS was selected for its comprehensive nature but required extensive preprocessing.

2. Research Question Definition:

- We had to avoid generic questions that were not actionable.
- Questions were aligned with dataset capability and statistical validity.
- Each question needed to support meaningful visualization.

Conclusion

Heart disease is influenced by an intricate combination of demographic, behavioral, clinical, and lifestyle factors. The dataset highlights how age, physical inactivity, smoking, poor general health, and stroke history significantly contribute to heart disease rates. While dietary habits play a role, they cannot independently mitigate risk when other strong predictors are present. The project also shows the importance of mental health, sleep patterns, and demographic inequalities in shaping overall health outcomes. This study provides the foundation for future machine-learning applications and advanced predictive modeling.

Appendix – Python EDA Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from scipy.spatial.distance import mahalanobis
from scipy import linalg
from pandas.plotting import scatter_matrix
```

1. Basic Overview

```
# Load data
df = pd.read_csv("merged_dataset_cleaned.csv")

print("\n--- Shape of Dataset ---")
print(df.shape)

print("\n--- Column Names ---")
print(df.columns.tolist())

print("\n--- Data Types ---")
print(df.dtypes)
```

```
print("\n--- Missing Values ---")
print(df.isnull().sum())
```

```
print("\n--- Duplicate Rows ---")
print(df.duplicated().sum())
```

2. Descriptive Statistics

```
print("\n--- Descriptive Statistics (Numeric) ---")
print(df.describe())
```

```
print("\n--- Descriptive Statistics (Categorical) ---")
print(df.describe(include='object'))
```

3. Categorical Variable Counts

```
categorical_cols = df.select_dtypes(include=['object', 'category']).columns
for col in categorical_cols:
    print(f"\n--- Value Counts: {col} ---")
    print(df[col].value_counts())
```

4. Univariate Distributions

```
num_cols = df.select_dtypes(include=[np.number]).columns.tolist()
cat_cols = df.select_dtypes(exclude=[np.number]).columns.tolist()
```

```
# ---- Numeric Columns ----
```

```
for col in num_cols:
    plt.figure(figsize=(10,4))
    plt.subplot(1,2,1)
    sns.histplot(df[col], kde=True, bins=30, color='teal')
    plt.title(f"Distribution of {col}")

    plt.subplot(1,2,2)
    sns.boxplot(x=df[col], color='orange')
```

```
plt.title(f"Boxplot of {col}")
plt.tight_layout()
plt.show()
```

---- Categorical Columns ----

```
for col in cat_cols:
    plt.figure(figsize=(8,4))
    df[col].value_counts().nlargest(10).plot(kind='bar', color='skyblue')
    plt.title(f"Top Categories in {col}")
    plt.xlabel(col)
    plt.ylabel("Count")
    plt.tight_layout()
    plt.show()
```

5. Bivariate Analysis

---- Numeric vs Numeric ----

```
for i, col1 in enumerate(num_cols):
    for col2 in num_cols[i+1:]:
        plt.figure(figsize=(6,4))
        sns.scatterplot(x=df[col1], y=df[col2], alpha=0.6, color='purple')
        plt.title(f"{col1} vs {col2}")
        plt.show()
```

---- Categorical vs Numeric ----

```
for cat in cat_cols:
    for num in num_cols:
        plt.figure(figsize=(8,4))
        sns.boxplot(x=cat, y=num, data=df)
        plt.title(f"{num} by {cat}")
        plt.xticks(rotation=45)
        plt.tight_layout()
        plt.show()
```

6. Multivariate

```
num_df = df.select_dtypes(include=['int64', 'float64'])
cols = num_df.columns # <-- FIXED: define cols here
```

```

if len(cols) >= 4:
    plt.figure(figsize=(8, 6))
    sns.scatterplot(
        data=num_df,
        x=cols[0],
        y=cols[1],
        hue=cols[2], # 3rd variable
        size=cols[3], # 4th variable
        sizes=(50, 250),
        alpha=0.7
    )
    plt.title(f"Multivariate Scatter Plot: {cols[0]}, {cols[1]}, {cols[2]}, {cols[3]}")
    plt.tight_layout()
    plt.show()
else:
    print("Need at least 4 numerical variables for multivariate scatterplot.")

```

7. Correlation Matrix (Heatmap)

Select only numerical columns

```

num_df = df.select_dtypes(include=['int64', 'float64'])

plt.figure(figsize=(10, 6))
sns.heatmap(num_df.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Matrix (Heatmap)")
plt.show()

```

References

1. CDC Behavioral Risk Factor Surveillance System (BRFSS)
2. Power BI Documentation
3. Public Health Literature on Cardiovascular Disease
4. Python Libraries: Pandas, Matplotlib